Mahmoud Ahmed Abdelawal

AI & Machine Learning Engineer

Contact Information

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Portfolio: http://mahmoudmahrous24.github.io/Portfolio

PROFESSIONAL SUMMARY

Military Status: Exempted

AI & Machine Learning Engineer and Computer Science graduate with hands-on experience developing end-to-end AI solutions across Computer Vision, Deep Learning, NLP, and Generative AI. Proficient in Python, TensorFlow, PyTorch, and FastAPI, with practical experience building multimodal AI systems, integrating machine learning models, and developing RAG-based applications. Key projects include a five-model AI-powered disaster management system and an award-winning clinical decision support platform. Strong analytical and problem-solving skills, with a focus on translating AI techniques into practical, real-world applications.

TECHNICAL SKILLS

Machine Learning & Deep Learning: Machine Learning, Deep Learning, Neural Networks, CNNs, GANs, Transfer Learning, Model Training

Computer Vision: YOLOv8, Segmentation, Image Classification, OpenCV, MiDaS Depth Estimation

NLP & Generative AI: Text Preprocessing, Tokenization, Embeddings, Sequence Modeling, RAG, ChromaDB, Vector Search, Gemini API

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-Learn, CLIP, BLIP, ConvNeXt

Programming Languages: Python, Java, C++, JavaScript

Data Science: EDA, Data Cleaning, Data Visualization, Pandas, NumPy, Matplotlib, Seaborn

Backend & DevOps: FastAPI, React, SQLite, Git, GitHub, Microsoft Azure, Docker, VS Code

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Soft Skills

Problem Solving, Analytical Thinking, Teamwork Communication, Adaptability and Eagerness to Learn

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EDUCATION

Bachelor of Computer Science and Information Technology
Egyptian E-Learning University (EELU), Sohag, Egypt (2022 : 2026)

Projects

S-Report AI: Universal Urban Intelligence & Disaster Management (Grade A+)

Project Link : https://github.com/MahmoudMahrous24/S-Report-AI-Vision

- Developed a hybrid multimodal AI system (graduation project) combining 5 models (CLIP, ConvNeXt-Tiny, BLIP, YOLOv8, MiDaS) for disaster classification.
- Integrated YOLOv8 segmentation and MiDaS depth estimation for spatial intelligence and scene understanding.
- Engineered a weighted decision and dispatch engine for automated emergency resource allocation.
- Architected a production-ready FastAPI backend with SQLite and Docker containerization.

DermaAssist AI — Clinical Decision Support System

Project Link: https://github.com/MahmoudMahrous24/AI-Hackathon-Sohag

- Built a bilingual (2 languages: Arabic/English) AI dermatology platform (1st place, AI Hackathon) grounded in WHO clinical guidelines.
- Implemented RAG pipeline using ChromaDB + Nomic Embeddings + Gemini Flash for evidence-based responses.
- Designed multimodal analysis across 3 input types (image, video, voice) and 2 user portals (Doctor/Patient) with prescription management.

Artificial Vincent Van Gogh

Project Link: https://github.com/DEPI-Project/Artificial-Vincent-Van-Gogh

- Created a generative AI system (DEPI Graduation Project) using DCGANs and neural style transfer to generate Van Gogh-inspired images.
- Trained deep learning models on a curated Van Gogh dataset using Python and GAN-based workflows.

Data Analysis Projects

Project Link: https://github.com/MahmoudMahrous24/Data-Analysis

- Conducted EDA, data cleaning, visualization, and insight extraction across real-world datasets.
- Constructed analytical pipelines using Python, Pandas, NumPy, Matplotlib, and Seaborn.

PROFESSIONAL EXPERIENCE

AI Data Trainer (Freelance/Contract) **2025-2026**

Crowdgen Platform

- Evaluated and trained AI models by providing high-quality data annotation and logical reasoning feedback.
- Ensured data integrity and algorithmic accuracy, developing strong analytical and problem-solving skills.

Languages

Arabic: Native
English: Professional Working Proficiency

CERTIFICATIONS & Courses

Data Analysis Diploma | Creativa Hub sohag | oct, nov 2026
Deep Learning, Computer Vision, and NLP | Hisham Asem | sep: dec 2025
Microsoft ML Engineer | Digital Egypt Pioneers Initiative (DEPI) | Mar 2024: nov 2024
English and Soft Skills | Digital Egypt Pioneers Initiative (DEPI) | Mar 2024: nov 2024
Huawei Hcia-AI V3.5 | sep: nov 2024

ACHIEVEMENTS

- Recognized by the **Minister of Communications of Egypt** as a top-performing DEPI student nationwide.
- Won 1st place at the AI Hackathon with DermaAssist AI (sponsored by the **Ministry of Communications**, **Orange Egypt**, **Instant**).